

SUNIX Linux Kernel Module

SDC Expansion Board Driver

Installation Guideline

Driver Ver. 2.1.1.0

Ver. 2.2

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1. Driver installation

- 1). Confirm the Kernel version is 4.11 or above, include 5.x and 6.x.

```
root@test-System-Product-Name: ~
root@test-System-Product-Name:~# uname -a
Linux test-System-Product-Name 6.5.0-45-generic #45-22.04.1-Ubuntu SMP PREEMPT_DYNAMIC
Mon Jul 15 16:40:02 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
root@test-System-Product-Name:~#
```

- 2). Confirm that the following packages have been installed:

- **make**
- **gcc**
- **gcc-12**
- **ncurses.h**

If these packages are not installed, please confirm Chapter 7 Troubleshooting.

- 3). Verify that the system detects the card by “**lspci**”:

\$lspci

```
root@test-System-Product-Name:~# lspci -v -s '0000:05:00.0'
05:00.0 Communication controller: SUNIX Co., Ltd. Device 2000 (rev 01)
Subsystem: SUNIX Co., Ltd. Device 0001
Flags: medium devsel, IRQ 255
Memory at 84c02000 (32-bit, non-prefetchable) [disabled] [size=2K]
I/O ports at 4000 [disabled] [size=64]
Memory at 84c00000 (32-bit, non-prefetchable) [disabled] [size=8K]
Capabilities: [40] Power Management version 3
```

- 4). Unzip “**SDCLinuxExpansionBoardDriver_VX.X.X.X.zip**”.

```
root@test-System-Product-Name:/home/test# ls
Desktop Documents Downloads Music Pictures Public SDCLinuxExpansionBoardDriver_ .zip snap Templates Videos
root@test-System-Product-Name:/home/test# unzip SDCLinuxExpansionBoardDriver_ .zip
Archive: SDCLinuxExpansionBoardDriver_ .zip
```

- 5). Enter “**./SDCLinuxExpansionBoardDriver_VX.X.X.X/driver**” directory.

Use “**make**” to build driver.

\$make

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# make
Compile SDC driver. (kernel 6.5.0-45-generic)

make CFLAGS_MODULE+=-DNETCAN CFLAGS_MODULE+=-Wframe-larger-than=2048 EXTRA_CFLAGS+="-DSDC_CREATE_UART=1 -DSDC_CREATE_DIO=1 -DSDC_CREATE_SPI=1
-DSDC_CREATE_CAN=1 -DSDC_CREATE_PARPORT=0" -C /lib/modules/6.5.0-45-generic/build M=/home/test/SDCLinuxExpansionBoardDriver_ /driver
modules
make[1]: Entering directory '/usr/src/linux-headers-6.5.0-45-generic'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04) 12.3.0
You are using: gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04) 12.3.0
```

GPIO optional

Select GPIO installation option instead of SDC Digital Driver.

➤ Unless the user's application uses “**gpiod**”, please skip this step.

\$make FSDC_ENABLE_GPIO=1

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# make FSDC_ENABLE_GPIO=1
```

6). Use “make install” to build and install driver.

\$make install

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# make install
Install SDC driver. (kernel 6.5.0-45-generic)

sudo cp -f ./obj/sunix-sdc.ko /lib/modules/6.5.0-45-generic/kernel/drivers/mfd
sudo cp -f ./obj/sdc_config.ko /lib/modules/6.5.0-45-generic/kernel/drivers/char
sudo cp -f ./obj/8250_sdc.ko /lib/modules/6.5.0-45-generic/kernel/drivers/tty/serial/8250
sudo cp -f ./obj/sdc_dio.ko /lib/modules/6.5.0-45-generic/kernel/drivers/char
sudo cp -f ./obj/spi-sdc.ko /lib/modules/6.5.0-45-generic/kernel/drivers/spi
sudo cp -f ./obj/sdc_mcp251x.ko /lib/modules/6.5.0-45-generic/kernel/drivers/net/can/spi
sudo cp -f ./obj/sdc_mcp251xfd.ko /lib/modules/6.5.0-45-generic/kernel/drivers/net/can/spi
sudo cp -f ./obj/can_sdc.ko /lib/modules/6.5.0-45-generic/kernel/drivers/net/can
sudo depmod -a
```

7). Reboot system, confirm the module is mounted

\$lspci -v -s “0000:xx:xx.x”

```
root@test-System-Product-Name:~# lspci -v -s '0000:05:00.0'
05:00.0 Communication controller: SUNIX Co., Ltd. Device 2000 (rev 01)
Subsystem: SUNIX Co., Ltd. Device 0001
Flags: medium devsel, IRQ 16
Memory at 84c02000 (32-bit, non-prefetchable) [size=2K]
I/O ports at 4000 [size=64]
Memory at 84c00000 (32-bit, non-prefetchable) [size=8K]
Capabilities: [40] Power Management version 3
Kernel driver in use: sunix_sdc
Kernel modules: sunix_sdc
```

8). Check the SDC device nodes *sdconfig_n* and *sdcdio_n* in */dev*.

```
root@test-System-Product-Name:~# ls /dev
autofs      drl          hidraw4      i2c-4      loop1      mem          nvme0n1p2   rtc          tty          tty19      tty3         tty40      tty51      tty62      ttyS14      ttyS25      ttyS8
block       drm_dp_aux0  hpet         i2c-5      loop2      queue       nvram        rtc0         tty0         tty2        tty30        tty41      tty52      tty63      ttyS15      ttyS26      ttyS9
btrfs-control ecryptfs    hugepages    i2c-6      loop3      mtd0         port         sdc_config0  tty1         tty20        tty31      tty42      tty53      tty7        ttyS16      ttyS27      udmabuf
bus          fb0          hwrng        i2c-7      loop4      mtd0ro       ppp          sdc_dio0     tty10        tty21        tty32      tty43      tty54      tty8        ttyS17      ttyS28      uhid
char         fd           i2c-0        i2c-8      loop5      mtd1         pps0         shm          tty11        tty22        tty33      tty44      tty55      tty9        ttyS18      ttyS29      uinput
console      full         i2c-1        i2c-9      loop6      mtdi         psaux        snapshot     tty12        tty23        tty34      tty45      tty56      ttyprintk  ttyS19      ttyS30      urandom
core         fuse         i2c-10       initctl    loop7      net          ptp0         snd          tty13        tty24        tty35      tty46      tty57      ttyS20      ttyS31      usb
cpu          gpiochip0    i2c-11       input      loop8      ng0n1        ptp0         stderr       tty14        tty25        tty36      tty47      tty58      ttyS21      ttyS32      userfaultfd
cpu_dma_latency hidraw0      i2c-12       kmsg       loop-control null          ptp1         stdio        tty15        tty26        tty37      tty48      tty59      ttyS22      ttyS33      userio
cuse         hidraw1      i2c-13       kvm         loop0      nvme0        pts          stdout       tty16        tty27        tty38      tty49      tty60      ttyS23      ttyS34      vcs
disk         hidraw2      i2c-2        log         loop1      nvme0n1      random       tpm0         tty17        tty28        tty39      tty50      tty61      ttyS24      ttyS35      vcs1
dma_heap     hidraw3      i2c-3        loop0       mei0       nvme0n1p1    rkill        tpmrm0       tty18        tty29        tty40      tty51      tty62      ttyS25      ttyS36      vcs2
```



GPIO optional

If user chooses to install the GPIO driver, *gpiochip_n* will be generated, while *sdcdio* will not be created.

```
root@test-System-Product-Name:~# ls /dev
autofs      drl          hidraw3      i2c-3      loop0      loop7      mtdi         psaux        snd          tty13      tty24      tty35      tty46      tty57      ttyS0      ttyS2      ttyS30      usb          vcsa2      vfio
block       drm_dp_aux0  hidraw4      i2c-4      loop1      loop8      net          ptp0         stderr       tty14      tty25      tty36      tty47      tty58      ttyS1      ttyS20      ttyS31      userfaultfd vcsa3      vga_arbiter
btrfs-control ecryptfs    hpet         i2c-5      loop2      loop9      ng0n1        ptp0         stdio        tty15      tty26      tty37      tty48      tty59      ttyS10      ttyS21      ttyS32      userio     vcsa4      vhost-net
bus          fb0          hugepages    i2c-6      loop3      loop-control null          ptp1         stdout       tty16      tty27      tty38      tty49      tty60      ttyS11      ttyS22      ttyS33      vcs5      vcsa5      vhost-vsock
char         fd           hwrng        i2c-7      loop4      loop12      napper       nvme0        pts          tty17      tty28      tty39      tty50      tty61      ttyS12      ttyS23      ttyS34      vcs6      vcsa6      zero
console      full         i2c-0        i2c-8      loop5      loop13      ncelog       nvme0n1      random       tty18      tty29      tty40      tty51      tty62      ttyS13      ttyS24      ttyS35      vcs7      vcsa7      zfs
core         fuse         i2c-1        i2c-9      loop6      loop14      mei0         nvme0n1p1    rkill        tty19      tty30      tty41      tty52      tty63      ttyS14      ttyS25      ttyS36      vcs8      vcsa8
cpu          gpiochip0    i2c-10       initctl    loop7      loop15      men          nvme0n1p2    rtc          tty20      tty31      tty42      tty53      tty64      ttyS15      ttyS26      ttyS37      vcs9      vcsa9
cpu_dma_latency hidraw0      i2c-11       input      loop8      loop16      queue       nvram        rtc0         tty21      tty32      tty43      tty54      tty65      ttyS16      ttyS27      ttyS38      vcs10     vcsa10
cuse         hidraw1      i2c-12       kmsg       loop-control mtd0         port         sdc_config0  tty22      tty33      tty44      tty55      tty66      ttyS17      ttyS28      ttyS39      vcs11     vcsa11
disk         hidraw2      i2c-13       kvm         loop0      mtd0ro       ppp          shm          tty23      tty34      tty45      tty56      tty67      ttyS18      ttyS29      ttyS40      vcs12     vcsa12
dma_heap     hidraw3      i2c-2        log         loop1      mtd1         pps0         snapshot     tty24      tty35      tty46      tty57      tty68      ttyS19      ttyS30      ttyS41      vcs13     vcsa13
```

2. Tool installation

- 1). Complete driver installation.
- 2). Enter “./SDCLinuxExpansionBoardDriverVX.X.X.X/tools” directory.
- 3). Enter the following commands in order.

```
$cp x86-64/sdc-config /usr/sbin or cp arm64/sdc-config /usr/sbin
$cp x86-64/sdc-uart-cfg /usr/sbin or cp arm64/sdc-uart-cfg /usr/sbin
```

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /tools# ls
arm64  x86-64
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /tools# cp x86-64/sdc-config /usr/sbin/
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /tools# cp x86-64/sdc-uart-cfg /usr/sbin/
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /tools#
```

4).



Ubuntu/Debian

Enter “./SDCLinuxExpansionBoardDriverVX.X.X.X/script/Ubuntu”

- If the user's system is UOS, simply select the /Ubuntu directory.

```
$cp sdc-board-cfg.sh /etc/init.d
$update-rc.d sdc-board-cfg.sh defaults
$/etc/init.d/sdc-board-cfg.sh start
$sdc-uart-cfg -s
```

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /script/Ubuntu# cp sdc-board-cfg.sh /etc/init.d/
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /script/Ubuntu# update-rc.d sdc-board-cfg.sh defaults
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /script/Ubuntu# /etc/init.d/sdc-board-cfg.sh start
Starting sdc-board-cfg.sh (via systemctl): sdc-board-cfg.service.
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /script/Ubuntu# sdc-uart-cfg -s
SAVE line=4;flags=00000000;
SAVE line=5;flags=00000000;
SAVE line=6;flags=00000013;
SAVE line=7;flags=00000013;
```

Red Hat Enterprise Linux/CentOS/Rocky

Enter “./SDCLinuxExpansionBoardDriverVX.X.X.X/script/RHEL”.

```
$cp sdc-start.sh /usr/local/bin/
$cp sdc-stop.sh /usr/local/bin/
$cp sdc-board-cfg.service /etc/systemd/system
$systemctl enable sdc-board-cfg
$ systemctl start sdc-board-cfg
$ sdc-uart-cfg -s
```

```
[root@localhost SDCLinuxExpansionBoardDriver_]# cd script/RHEL/
[root@localhost RHEL]# cp sdc-start.sh /usr/local/bin/
[root@localhost RHEL]# cp sdc-stop.sh /usr/local/bin/
[root@localhost RHEL]# cp sdc-board-cfg.service /etc/systemd/system
[root@localhost RHEL]# systemctl enable sdc-board-cfg
Created symlink /etc/systemd/system/multi-user.target.wants/sdc-board-cfg.service → /etc/systemd/system/sdc-board-cfg.service.
[root@localhost RHEL]# systemctl start sdc-board-cfg
[root@localhost RHEL]# sdc-uart-cfg -s
SAVE line=1;flags=00000000;
SAVE line=2;flags=00000000;
SAVE line=3;flags=00000013;
```

2. Terminal tool snxterm installation

- 1). Enter “./SDCLinuxExpansionBoardDriverVX.X.X.X/snxterm” directory.
- 2). Use “make install” to build and install snxterm.

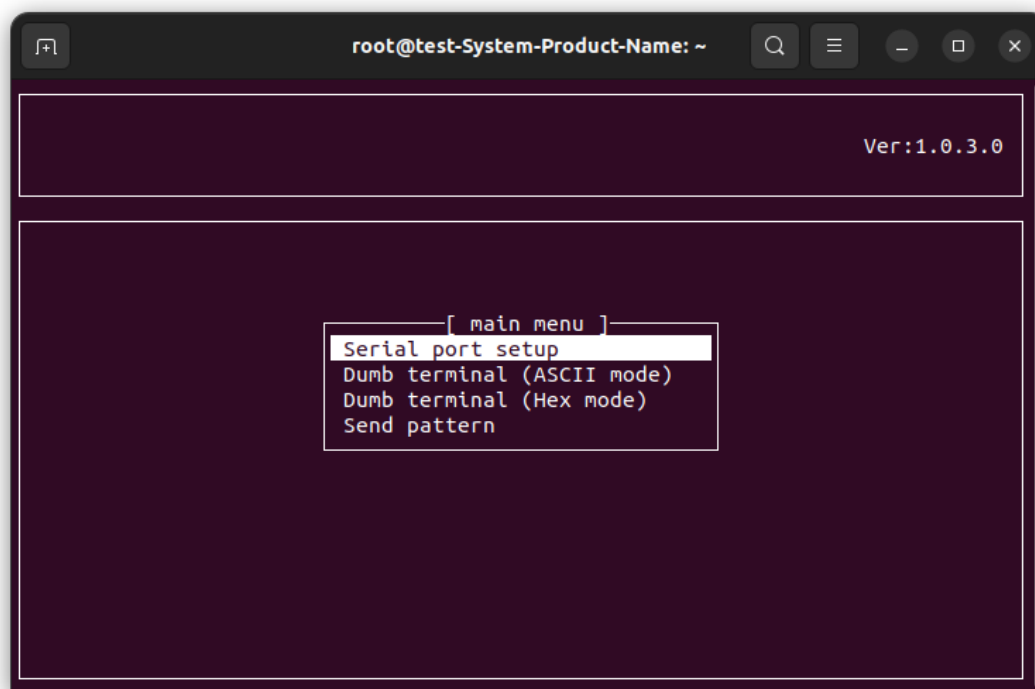
\$make install

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /snxterm# make install
gcc -O2 -pthread -pipe -o snxterm snxterm.c -lncurses
cp -p snxterm /usr/bin
```

3). **snxterm** is a terminal tool, the tool could test serial port communication, Enter “**snxterm**” to execute the application.

- If the user is not using an SDC card with a serial port, this installation can be skipped.

\$snxterm



3. Driver uninstallation

- 1). Enter “./SDCLinuxExpansionBoardDriverVX.X.X.X/driver” directory.
- 2). Close all driver device node (/dev/sdc_dio0, /dev/ttyS4... etc.).
- 3). Use “make uninstall” to uninstall driver.

\$make uninstall

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# make uninstall
Uninstall SDC driver. (kernel 6.5.0-45-generic)

sudo rm -f /lib/modules/6.5.0-45-generic/kernel/drivers/mfd/sunix-sdc.ko
sudo rm -f /lib/modules/6.5.0-45-generic/kernel/drivers/char/sdc_config.ko
sudo rm -f /lib/modules/6.5.0-45-generic/kernel/drivers/tty/serial/8250/8250_sdc.ko
sudo rm -f /lib/modules/6.5.0-45-generic/kernel/drivers/char/sdc_dio.ko
sudo rm -f /lib/modules/6.5.0-45-generic/kernel/drivers/spi/spi-sdc.ko
sudo rm -f /lib/modules/6.5.0-45-generic/kernel/drivers/net/can/spi/sdc_mcp251x.ko
sudo rm -f /lib/modules/6.5.0-45-generic/kernel/drivers/net/can/spi/sdc_mcp251xfd.ko
sudo rm -f /lib/modules/6.5.0-45-generic/kernel/drivers/net/can/can_sdc.ko
sudo depmod -a
```

- 4). Use “make clean” to remove all built files.

\$make clean

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# make clean
Clean SDC driver. (kernel 6.5.0-45-generic)

make -C /lib/modules/6.5.0-45-generic/build M=/home/test/SDCLinuxExpansionBoardDriver_ /driver clean
make[1]: Entering directory '/usr/src/linux-headers-6.5.0-45-generic'
  CLEAN  /home/test/SDCLinuxExpansionBoardDriver_V2.1.0.0/driver/Module.symvers
make[1]: Leaving directory '/usr/src/linux-headers-6.5.0-45-generic'
rm -Rf .tmp_versions
rm -f .cache.*
```

4. Tools uninstallation

- 1). Complete driver uninstallation.
- 2). Enter the following commands in order to uninstall tools.

 **Ubuntu/Debian**

```
$/etc/init.d/sdc-board-cfg.sh stop
$update-rc.d -f sdc-board-cfg.sh remove
$rm /etc/init.d/sdc-board-cfg.sh
$rm -Rf /usr/share/sdc_8250/
$rm /usr/sbin/sdc-uart-cfg
$rm /usr/sbin/sdc-config
```

```
root@test-System-Product-Name:~# /etc/init.d/sdc-board-cfg.sh stop
Stopping sdc-board-cfg.sh (via systemctl): sdc-board-cfg.service.
root@test-System-Product-Name:~# update-rc.d -f sdc-board-cfg.sh remove
root@test-System-Product-Name:~# rm /etc/init.d/sdc-board-cfg.sh
root@test-System-Product-Name:~# rm -Rf /usr/share/sdc_8250/
```

```
root@test-System-Product-Name:~# rm /usr/sbin/sdc-config
root@test-System-Product-Name:~# rm /usr/sbin/sdc-uart-cfg
```

 Red Hat Enterprise Linux/CentOS/Rocky

```
$systemctl stop sdc-board-cfg
$systemctl disable sdc-board-cfg
$rm /usr/local/bin/sdc-start.sh
$rm /usr/local/bin/sdc-stop.sh
$rm /etc/systemd/system/sdc-board-cfg.service
$rm -Rf /usr/share/sdc_8250/
$rm /usr/sbin/sdc-uart-cfg
$rm /usr/sbin/sdc-config
```

```
[root@localhost test]# systemctl stop sdc-board-cfg
[root@localhost test]# systemctl disable sdc-board-cfg
Removed "/etc/systemd/system/multi-user.target.wants/sdc-board-cfg.service".
[root@localhost test]# rm /usr/local/bin/sdc-start.sh
rm: remove regular file '/usr/local/bin/sdc-start.sh'? y
[root@localhost test]# rm /usr/local/bin/sdc-stop.sh
rm: remove regular file '/usr/local/bin/sdc-stop.sh'? y
[root@localhost test]# rm /etc/systemd/system/sdc-board-cfg.service
rm: remove regular file '/etc/systemd/system/sdc-board-cfg.service'? y
[root@localhost test]# rm -Rf /usr/share/sdc_8250/
[root@localhost test]#
```

```
[root@localhost test]# rm /usr/sbin/sdc-config
rm: remove regular file '/usr/sbin/sdc-config'? y
[root@localhost test]# rm /usr/sbin/sdc-uart-cfg
rm: remove regular file '/usr/sbin/sdc-uart-cfg'? y
```

➤ In RHEL, a reboot is required for the uninstallation to take effect.

5. Terminal tool snxterm uninstallation

- 1). Enter “./SDCLinuxExpansionBoardDriverVX.X.X.X/snxterm” directory.
- 2). Complete tools uninstallation.
- 3). Enter the following command to uninstall **snxterm**.

```
$make clean
```

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /snxterm# make clean
rm -f snxterm
rm -f *~ *.o
rm -f /usr/bin/snxterm
```

6. Troubleshooting

Errors often occur when compiling source code packages. The following is a compilation of some common errors.

1). “make” not found

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# make
Command 'make' not found, but can be installed with:
apt install make # version 4.3-4.1build1, or
apt install make-guile # version 4.3-4.1build1
```

Ubuntu command “apt-get install make-guile”

```
$apt-get install make-guile
```

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# apt-get install make-guile
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
make-doc
The following NEW packages will be installed:
make-guile
```

2). gcc-12 not found

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# make
Compile SDC driver. (kernel 6.5.0-45-generic)
make CFLAGS_MODULE+=-DNETCAN CFLAGS_MODULE+=-Wframe-larger-than=2048 EXTRA_CFLAGS+="-DSDC_CREATE_UART=1 -DSDC_CREATE_DIO=1 -DSDC_CREATE_SPI=1
-DSDC_CREATE_CAN=1 -DSDC_CREATE_PARPORT=0" -C /lib/modules/6.5.0-45-generic/build M=/home/test/SDCLinuxExpansionBoardDriver_ /driver
modules
make[1]: Entering directory '/usr/src/linux-headers-6.5.0-45-generic'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1-22.04) 12.3.0
You are using:
CC [M] /home/test/SDCLinuxExpansionBoardDriver_ /driver/sunix-sdc.o
/bin/sh: 1: gcc-12: not found
make[3]: *** [scripts/Makefile.build:251: /home/test/SDCLinuxExpansionBoardDriver_ /driver/sunix-sdc.o] Error 127
make[2]: *** [/usr/src/linux-headers-6.5.0-45-generic/Makefile:2039: /home/test/SDCLinuxExpansionBoardDriver_ /driver] Error 2
make[1]: *** [Makefile:234: _sub-make] Error 2
make[1]: Leaving directory '/usr/src/linux-headers-6.5.0-45-generic'
make: *** [Makefile:70: default] Error 2
```

Ubuntu command “apt-get install gcc-12”

```
$apt-get install gcc-12
```

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /driver# apt-get install gcc-12
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu cpp-12 libasan8 libbinutils libc-dev-bin libc-devtools libc6-dev
  libnsd-dev libtirpc-dev libtsan2 libubsan1 linux-libc-dev manpages-dev rpcsvc-proto
```

3). “snxterm” installation, gcc not found

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /snxterm# make
gcc -O2 -pthread -pipe -o snxterm snxterm.c -lncurses
make: gcc: No such file or directory
make: *** [Makefile:13: snxterm] Error 127
```

Ubuntu command “apt-get install gcc”

```
$apt-get install gcc
```

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /snxterm# apt-get install gcc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu cpp-11 gcc-11 gcc-11-base libasan6 libbinutils libc-dev
  libgcc-11-dev libitm1 liblsan0 libnsd-dev libtirpc-dev libtsan0 libubsan1 linux-libc-dev manpages-dev rpcs
Suggested packages:
  binutils-doc gcc-11-locales gcc-multilib autoconf automake libtool flex bison gcc-doc gcc-11-multilib gcc-
Recommended packages:
  libnss-nis libnss-nisplus
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu gcc gcc-11 libasan6 libbinutils libc-dev-bin libc-devto
  libnsd-dev libtirpc-dev libtsan0 libubsan1 linux-libc-dev manpages-dev rpcsvc-proto
The following packages will be upgraded:
  cpp-11 gcc-11-base libc6 libc6-dbg
4 upgraded, 24 newly installed, 0 to remove and 437 not upgraded.
Need to get 39.0 MB/66.0 MB of archives.
After this operation, 131 MB of additional disk space will be used.
```

4). “snxterm” installation, ncurses.h: no such file

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /snxterm# make
gcc -O2 -pthread -pipe -o snxterm snxterm.c -lncurses
snxterm.c:12:10: fatal error: ncurses.h: No such file or directory
 12 | #include <ncurses.h>
    |          ^
compilation terminated.
make: *** [Makefile:13: snxterm] Error 1
```

Ubuntu command “apt-get install libncurses5-dev”

```
$apt-get install libncurses5-dev
```

RHEL command “yum install ncurses-devel”

\$yum install ncurses-devel

```
root@test-System-Product-Name:/home/test/SDCLinuxExpansionBoardDriver_ /snxterm# apt-get install libncurses5-dev
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libncurses-dev
Suggested packages:
  ncurses-doc
The following NEW packages will be installed:
  libncurses-dev libncurses5-dev
```